

# A Four-Terminal Planar Lower Bound and a Sharp Equal-Cost Gadget for Cost-Preserving Unsplittable-Flow Rounding

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This is an unrefereed research disclosure. Its mathematical claim is deliberately narrower than “the exact constant”: it gives a lower bound for an existential, upper-only, cost-preserving planar rounding constant and a sharp theorem only inside an explicitly restricted model. Novelty searches are ongoing and are not exhaustive.

## Abstract

For a planar acyclic single-source unsplittable-flow instance with fractional arc-load vector  $x$ , maximum demand  $d_{\max}$ , and nonnegative arc-cost vector  $c$ , let  $\alpha_{\text{pl}}$  be the least universal factor for which there always exists an unsplittable routing  $y$  satisfying  $c^\top y \leq c^\top x$  and  $y(a) \leq x(a) + \alpha_{\text{pl}} d_{\max}$  on every arc. We give a four-terminal planar family proving

$$\alpha_{\text{pl}} \geq L := \frac{299 - 41\sqrt{41}}{32} = 1.1397470789\dots$$

The bound is a supremal family bound: for every  $\eta > 0$  we give a rational instance requiring an overload larger than  $(L - \eta)d_{\max}$ . We also give an attained integer certificate of  $335/294 = 1.139455782312\dots$ . Finally, we prove that  $L$  is the exact supremum for the fixed four-terminal topology when every full expensive choice has equal cost and cost feasibility forces at least two cheap choices. The known planar theorem gives the algorithmic, two-sided upper bound  $\alpha_{\text{pl}} \leq 2$ .

## 1 Definitions and scope

Let  $G = (V, A)$  be a finite directed graph, let  $s \in V$  be a source, and let  $T \subseteq V \setminus \{s\}$  be terminals with demands  $d = (d_t)_{t \in T} \in \mathbb{Q}_{\geq 0}^T$ . A splittable flow may be specified by nonnegative path amounts  $\lambda_{t,P}$  satisfying

$$\sum_{P \text{ an } s\text{-}t \text{ path}} \lambda_{t,P} = d_t \quad (t \in T).$$

Its aggregate arc loads are

$$x(a) = \sum_{t \in T} \sum_{P \ni a} \lambda_{t,P}.$$

An unsplittable routing  $\mathcal{P} = (P_t)_{t \in T}$  chooses one  $s$ - $t$  path for every terminal, and has aggregate loads

$$\text{flow}_{\mathcal{P}}(a) = \sum_{t \in T: a \in P_t} d_t.$$

For  $c \in \mathbb{Q}_{\geq 0}^A$ , define  $\text{cost}_c(z) := \sum_{a \in A} c(a)z(a)$  and  $d_{\max} := \max_{t \in T} d_t$ .

Following the planar convention in Traub–Vargas Koch–Zenklusen, call  $(G, s, T, d, x)$  a planar SSUF instance when  $G$  is acyclic and planar. Define

$$\alpha_{\text{pl}} := \inf \left\{ \alpha \geq 0 : \text{for every planar SSUF instance } (G, s, T, d, x) \text{ and every } c \in \mathbb{Q}_{\geq 0}^A, \right. \\ \left. \text{there exists an unsplittable } \mathcal{P} \text{ such that } \text{cost}_c(\text{flow}_{\mathcal{P}}) \leq \text{cost}_c(x) \right. \\ \left. \text{and } \text{flow}_{\mathcal{P}}(a) \leq x(a) + \alpha d_{\max} \text{ for every } a \in A \right\}.$$

This is an *existential* and *upper-only* constant. It is distinct from an algorithmic constant and from the cost-free, two-sided discrepancy question of Morell and Skutella. The algorithmic theorem of Traub–Vargas Koch–Zenklusen gives the stronger two-sided estimate  $|\text{flow}_{\mathcal{P}}(a) - x(a)| \leq 2d_{\max}$  while preserving cost, and hence implies  $\alpha_{\text{pl}} \leq 2$ .

## 2 The fixed planar digraph

Let

$$V = \{s = v_0, v_1, v_2, v_3, v_4, v_5, t_1, t_2, t_3, t_4\}.$$

For  $j \in \{1, \dots, 5\}$  write  $a_j = (v_{j-1}, v_j)$ . The arc set is

$$A = \{a_j : 1 \leq j \leq 5\} \\ \cup \{(s, t_1), (v_3, t_1), (s, t_2), (v_5, t_2), (v_1, t_3), (v_5, t_3), (v_2, t_4), (v_4, t_4)\}.$$

These are all the arcs; in particular, every  $t_i$  is a sink.

Each terminal has two designated paths:

$$\begin{array}{ll} P_1^E = (s, t_1), & P_1^C = (s, v_1, v_2, v_3, t_1), \\ P_2^E = (s, t_2), & P_2^C = (s, v_1, v_2, v_3, v_4, v_5, t_2), \\ P_3^E = (s, v_1, t_3), & P_3^C = (s, v_1, v_2, v_3, v_4, v_5, t_3), \\ P_4^E = (s, v_1, v_2, t_4), & P_4^C = (s, v_1, v_2, v_3, v_4, t_4). \end{array}$$

The superscripts E and C will stand for expensive and cheap once costs are introduced.

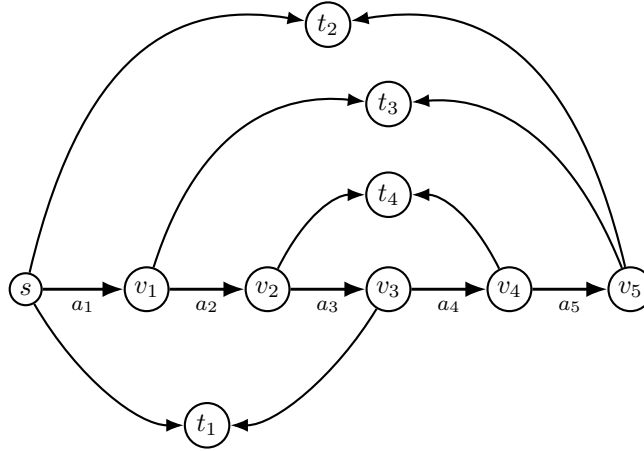


Figure 1: The fixed digraph. The edge list, rather than the drawing, is the formal specification.

**Lemma 1.** *The graph  $G$  is a planar DAG, and  $P_i^E, P_i^C$  are exactly the two  $s$ - $t_i$  paths for each  $i \in \{1, 2, 3, 4\}$ .*

*Proof.* The order  $s, v_1, v_2, v_3, v_4, v_5, t_1, t_2, t_3, t_4$  is topological. A plane embedding is shown in figure 1; equivalently, a clockwise rotation system is

$$\begin{aligned} s : (v_1, t_2, t_1), \quad v_1 : (s, v_2, t_3), \quad v_2 : (v_1, v_3, t_4), \quad v_3 : (v_2, t_1, v_4), \\ v_4 : (v_3, v_5, t_4), \quad v_5 : (v_4, t_2, t_3), \quad t_1 : (v_3, s), \quad t_2 : (v_5, s), \quad t_3 : (v_5, v_1), \quad t_4 : (v_4, v_2). \end{aligned}$$

Face tracing gives five faces, and  $|V| - |A| + 5 = 10 - 13 + 5 = 2$ . Finally, every terminal has precisely the two listed in-neighbors, and the trunk gives the unique path from  $s$  to each such predecessor. Thus there are exactly two source-terminal paths.  $\square$

The sets used in the discrepancy calculation are not the complete cheap paths. They are the supports on the trunk of the path differences:

$$I_i := \{a_j : \mathbf{1}_{P_i^C}(a_j) - \mathbf{1}_{P_i^E}(a_j) = 1\}.$$

Explicitly,

$$I_1 = \{a_1, a_2, a_3\}, \quad I_2 = \{a_1, a_2, a_3, a_4, a_5\}, \quad I_3 = \{a_2, a_3, a_4, a_5\}, \quad I_4 = \{a_3, a_4\}.$$

For example, both paths to  $t_3$  use  $a_1$ , so  $a_1 \notin I_3$ .

*Remark 2* (The graph is outside the series-parallel positive class). The underlying undirected graph has a  $K_4$  minor with connected branch sets

$$\{s, t_1, t_2\}, \quad \{v_1, v_2, t_3, t_4\}, \quad \{v_3, v_4\}, \quad \{v_5\}.$$

Every two branch sets are adjacent. In particular, the underlying undirected graph is not two-terminal series-parallel, since such graphs are  $K_4$ -minor-free. Thus the example lies outside the recursively constructed series-parallel digraph class used by Almoghrabi–Skutella–Warode and does not conflict with their strict  $d_{\max}$  deviation theorem.

### 3 The rational parametric family

Fix

$$q \in (\sqrt{3} - 1, 1) \cap \mathbb{Q}, \quad \varepsilon \in \mathbb{Q}, \quad 0 < \varepsilon < 3 - 2q - q^2.$$

Set

$$(d_1, d_2, d_3, d_4) = (1, q^2, q, 1)$$

and route the fraction  $p_i$  of terminal  $i$ 's demand on  $P_i^C$ , where

$$p_1 = 1 - q^2, \quad p_2 = q^2 + 2q - 2 + \varepsilon, \quad p_3 = p_4 = 1 - q.$$

The parameter restrictions imply  $0 < p_i < 1$  for every  $i$ . Indeed,  $q > \sqrt{3} - 1$  gives  $q^2 + 2q - 2 > 0$ , while the upper bound on  $\varepsilon$  gives  $p_2 < 1$ . Also

$$\sum_{i=1}^4 p_i = 1 + \varepsilon, \quad d_{\max} = 1.$$

Costs are specified on arcs, not assigned abstractly to paths. Put

$$c(s, t_1) = 1, \quad c(s, t_2) = \frac{1}{q^2}, \quad c(v_1, t_3) = \frac{1}{q}, \quad c(v_2, t_4) = 1,$$

and put cost zero on every other arc. These four charged arcs occur only on the corresponding expensive paths. Consequently

$$c(P_i^E) = \frac{1}{d_i}, \quad c(P_i^C) = 0,$$

so routing all of demand  $d_i$  expensively costs one.

Let  $x$  be the splittable flow that sends  $d_i p_i$  on  $P_i^C$  and  $d_i(1 - p_i)$  on  $P_i^E$ .

**Lemma 3** (Cost forcing). *The fractional cost is  $\text{cost}_c(x) = 3 - \varepsilon$ . Every unsplittable routing of cost at most  $\text{cost}_c(x)$  chooses at least two cheap paths.*

*Proof.* Terminal  $i$  contributes

$$d_i(1 - p_i)c(P_i^E) = d_i(1 - p_i)\frac{1}{d_i} = 1 - p_i$$

to the fractional cost. Hence  $\text{cost}_c(x) = \sum_i (1 - p_i) = 3 - \varepsilon$ . An unsplittable routing has cost equal to its integer number of expensive choices. Since  $3 - \varepsilon < 3$ , a cost-nonincreasing routing uses at most two expensive paths and therefore at least two cheap paths.  $\square$

For a cheap-set  $S \subseteq \{1, 2, 3, 4\}$ , let  $y^S$  denote the unsplittable routing that uses  $P_i^C$  exactly for  $i \in S$ .

**Lemma 4** (Monotonicity on the trunk). *If  $S \subseteq S'$ , then*

$$\text{flow}_{y^{S'}}(a) - x(a) \geq \text{flow}_{y^S}(a) - x(a) \quad \text{for every trunk arc } a.$$

*In particular, every routing with at least three cheap choices coordinatewise dominates, on the trunk, a routing with exactly two cheap choices.*

*Proof.* Switching terminal  $i$  from  $P_i^E$  to  $P_i^C$  changes the trunk load by  $d_i \mathbf{1}_{I_i}$ , which is coordinatewise nonnegative. Apply such switches one at a time.  $\square$

On an arc of  $I_i$ , the overload contribution relative to  $x$  is  $d_i(1 - p_i)$  when  $i$  is cheap and  $-d_i p_i$  when  $i$  is expensive. These values are

| $i$ | cheap contribution                | expensive contribution             |
|-----|-----------------------------------|------------------------------------|
| 1   | $q^2$                             | $q^2 - 1$                          |
| 2   | $q^2(3 - q^2 - 2q - \varepsilon)$ | $-q^2(q^2 + 2q - 2 + \varepsilon)$ |
| 3   | $q^2$                             | $q^2 - q$                          |
| 4   | $q$                               | $q - 1$                            |

Define

$$R(q, \varepsilon) := q^2(4 - q^2 - 2q - \varepsilon).$$

For the six exactly-two-cheap routings, direct substitution gives:

| $S$        | witness arc | $\text{flow}_{y^S}(a) - x(a)$ |
|------------|-------------|-------------------------------|
| $\{1, 2\}$ | $a_1$       | $R(q, \varepsilon)$           |
| $\{1, 3\}$ | $a_2$       | $R(q, \varepsilon)$           |
| $\{1, 4\}$ | $a_3$       | $R(q, \varepsilon)$           |
| $\{2, 3\}$ | $a_5$       | $R(q, \varepsilon)$           |
| $\{2, 4\}$ | $a_4$       | $R(q, \varepsilon)$           |
| $\{3, 4\}$ | $a_4$       | $R(q, \varepsilon) + q - q^2$ |

Since  $q - q^2 > 0$ , every row is at least  $R(q, \varepsilon)$ .

**Proposition 5.** *For every admissible rational pair  $(q, \varepsilon)$ , every cost-nonincreasing unsplittable routing overloads some arc by at least  $R(q, \varepsilon)d_{\max} = R(q, \varepsilon)$ .*

*Proof.* By lemma 3, every cost-nonincreasing routing has at least two cheap choices. If it has exactly two, use the table above. If it has more, use lemma 4 and retain any two of its cheap choices.  $\square$

**Theorem 6.**

$$\alpha_{\text{pl}} \geq L := \frac{299 - 41\sqrt{41}}{32} = 1.139747070789\dots$$

*More precisely, for every  $\eta > 0$  there is a rational four-terminal planar SSUF instance for which every cost-nonincreasing unsplittable routing overloads some arc by more than  $(L - \eta)d_{\max}$ .*

*Proof.* At  $\varepsilon = 0$ , write

$$f(q) := R(q, 0) = q^2(4 - q^2 - 2q).$$

Then

$$f'(q) = 2q(4 - 3q - 2q^2).$$

The unique maximizer in  $(\sqrt{3} - 1, 1)$  is

$$q_* = \frac{\sqrt{41} - 3}{4},$$

and direct substitution gives

$$f(q_*) = \frac{299 - 41\sqrt{41}}{32} = L.$$

Fix  $\eta > 0$ . By continuity of  $f$  and density of the rationals, choose  $q \in (\sqrt{3} - 1, 1) \cap \mathbb{Q}$  such that  $f(q) > L - \eta/2$ . Then choose rational  $\varepsilon$  satisfying

$$0 < \varepsilon < \min \left\{ 3 - 2q - q^2, \frac{\eta}{2q^2} \right\}.$$

The instance is admissible, all of its data are rational, and

$$R(q, \varepsilon) = f(q) - q^2\varepsilon > L - \eta.$$

Now apply proposition 5. Since this holds for every  $\eta > 0$ , no universal factor below  $L$  can satisfy the definition of  $\alpha_{\text{pl}}$ .  $\square$

**Corollary 7.**

$$\frac{299 - 41\sqrt{41}}{32} \leq \alpha_{\text{pl}} \leq 2.$$

*Proof.* The lower bound is theorem 6. The upper bound follows from the cost-preserving two-sided  $2d_{\max}$  planar rounding theorem of Traub–Vargas Koch–Zenklusen.  $\square$

## 4 Sharpness within an equal-full-cost two-cheap model

The construction above is tuned, but not arbitrary. In this section we show that it is optimal within a precise fixed-topology model.

Keep the digraph and the two paths per terminal from figure 1. The real relaxation of the model consists of demands  $d \in (0, 1]^4$  with  $\max_i d_i = 1$ , cheap fractions  $p \in [0, 1]^4$ , and a number  $K > 0$  satisfying

$$1 < \sum_{i=1}^4 p_i \leq 2.$$

Give every cheap path cost zero by charging only the private arc of the corresponding expensive path, at per-unit cost  $K/d_i$ . Thus every full expensive choice costs  $K$ . The fractional cost is  $K(4 - \sum_i p_i) \in [2K, 3K)$ , so a routing is cost-nonincreasing exactly when it has at least two cheap choices.

For these real data let  $x$  send  $d_i p_i$  cheaply and  $d_i(1 - p_i)$  expensively, and let  $\beta_G^{\text{eq},2}$  be the supremum of

$$\min_{S \subseteq \{1,2,3,4\}: |S| \geq 2} \max_{a \in A} (\text{flow}_{y^S}(a) - x(a)).$$

The superscript records the equal-full-cost and two-cheap regime. This is a restricted fixed-topology constant; it does not permit unequal full-route costs. Write  $\beta_{G,\mathbb{Q}}^{\text{eq},2}$  for the same supremum with  $d, p \in \mathbb{Q}^4$  (and, without loss of generality, rational  $K > 0$ ).

**Theorem 8** (Sharpness in the restricted model).

$$\beta_G^{\text{eq},2} = \beta_{G,\mathbb{Q}}^{\text{eq},2} = \frac{299 - 41\sqrt{41}}{32} = L.$$

*Proof.* The family in proposition 5 and theorem 6 has equal full expensive-route costs and  $\sum_i p_i = 1 + \varepsilon \in (1, 2)$ , so it proves  $\beta_G^{\text{eq},2} \geq L$ . We prove the reverse inequality.

Fix arbitrary data in the restricted model. Decrease the cheap fractions coordinatewise, if necessary, to numbers  $\hat{p}_i \leq p_i$  satisfying  $\sum_i \hat{p}_i = 1$ . For every exactly-two-cheap routing and every trunk arc, this change can only increase the deviation from the fractional flow: for a cheap terminal the contribution changes from  $d_i(1 - p_i)$  to  $d_i(1 - \hat{p}_i)$ , and for an expensive terminal it changes from  $-d_i p_i$  to  $-d_i \hat{p}_i$ . Thus an upper bound for the boundary data  $\hat{p}$  is also an upper bound for the original data. Every exactly-two-cheap routing is cost-feasible in the original instance.

Write

$$\ell_i := d_i \hat{p}_i, \quad e_i := d_i(1 - \hat{p}_i).$$

Positive deviations on private terminal arcs are at most  $d_i \leq 1$ , so only trunk deviations can matter for a bound larger than one. Let  $M_{ij}$  be the maximum trunk deviation when precisely terminals  $i$  and  $j$  are cheap. Directly from the four difference supports  $I_i$ ,

| $\{i, j\}$ | $M_{ij}$                                            |
|------------|-----------------------------------------------------|
| $\{1, 2\}$ | $e_1 + e_2$                                         |
| $\{1, 3\}$ | $e_1 + e_3 - \ell_2$                                |
| $\{1, 4\}$ | $\max\{e_1 - \ell_2, e_1 + e_4 - \ell_2 - \ell_3\}$ |
| $\{2, 3\}$ | $e_2 + e_3$                                         |
| $\{2, 4\}$ | $\max\{e_2 - \ell_1, e_2 + e_4 - \ell_3\}$          |
| $\{3, 4\}$ | $e_3 + e_4 - \ell_2$                                |

If  $e_4 < \ell_3$ , then  $M_{14} = e_1 - \ell_2 \leq 1$ , and we are done. Hence assume  $e_4 \geq \ell_3$ . In that case

$$M_{14} = e_1 + e_4 - \ell_2 - \ell_3, \quad M_{24} = e_2 + e_4 - \ell_3.$$

Because  $d_1, d_4 \leq 1$ ,

$$\widehat{p}_1 = 1 - \frac{e_1}{d_1} \leq 1 - e_1, \quad \widehat{p}_4 \leq 1 - e_4.$$

Using  $\sum_i \widehat{p}_i = 1$  gives

$$e_1 + e_4 \leq 1 + \widehat{p}_2 + \widehat{p}_3. \quad (1)$$

Put

$$s := d_2, \quad t := d_3, \quad p := \widehat{p}_2, \quad q := \widehat{p}_3.$$

Thus  $0 < s, t \leq 1$ ,  $p, q \geq 0$ , and  $p + q \leq 1$ , while

$$e_2 = s(1 - p), \quad \ell_2 = sp, \quad e_3 = t(1 - q), \quad \ell_3 = tq.$$

Define

$$L_0 := \min\{e_2, e_3 - \ell_2\}, \quad R_0 := \min\{e_2 - \ell_3, e_3 - \ell_2\}.$$

The first two rows involving terminal 1 have minimum  $e_1 + L_0$ , while the last two rows involving terminal 4 have minimum  $e_4 + R_0$ . Therefore, using (1),

$$\min_{|S|=2} M_S \leq \min\{T_1, T_2, T_3\}, \quad (2)$$

where

$$T_1 := e_2 + e_3, \quad T_2 := 1 + p + q - \ell_2 - \ell_3, \quad T_3 := \frac{1 + p + q + L_0 + R_0}{2}.$$

Indeed,  $T_1 = M_{23}$ ,  $M_{14} \leq T_2$ , and the minimum of  $e_1 + L_0$  and  $e_4 + R_0$  is at most their average.

First suppose  $t \geq s$ . The difference between the two candidates in  $R_0$  is

$$(e_3 - \ell_2) - (e_2 - \ell_3) = t - s,$$

so  $R_0 = e_2 - \ell_3$ . Also,  $L_0 = e_2$  when  $e_3 \geq s$ , and  $L_0 = e_3 - \ell_2$  when  $e_3 \leq s$ . Consequently  $T_3$  is at most

$$\widetilde{T}_3 := \frac{1 + 2s + (1 - 2s)p + (1 - t)q}{2}.$$

If  $e_3 \geq s$  then equality holds; if  $e_3 \leq s$ , the difference  $\widetilde{T}_3 - T_3$  is  $(s - e_3)/2 \geq 0$ .

If  $s + s/t \leq 1$ , then  $s \leq 1/2$  and

$$(1 - 2s)T_1 + 2s\widetilde{T}_3 = t + 2s(1 - t) + q(s(1 + t) - t) \leq 1.$$

Indeed,  $s \leq t/(1 + t)$  makes the coefficient of  $q$  nonpositive, while

$$t + 2s(1 - t) \leq t + \frac{2t(1 - t)}{1 + t} = \frac{t(3 - t)}{1 + t} \leq 1.$$

The coefficients on the left are nonnegative and sum to one, so (2) is at most one.

Now assume  $s + s/t \geq 1$ . Set

$$\lambda_1 := \frac{s}{t} - s, \quad \lambda_2 := s + \frac{s}{t} - 1, \quad \lambda_3 := 2 \left(1 - \frac{s}{t}\right).$$

These numbers are nonnegative and sum to one. A direct cancellation gives

$$\lambda_1 T_1 + \lambda_2 T_2 + \lambda_3 \tilde{T}_3 = g(s, t) := s \left( 4 - s - t - \frac{s}{t} \right).$$

Hence the right-hand side of (2) is at most  $g(s, t)$ .

It remains to consider  $t \leq s$ . Now both minima in  $L_0$  and  $R_0$  select  $e_3 - \ell_2$ , and hence

$$T_3 = \frac{1 + 2t + (1 - 2s)p + (1 - 2t)q}{2}.$$

If  $t \leq 1/2$ , then

$$(1 - 2t)T_1 + 2tT_3 = s + 2t - 2st + p(t - s) \leq 1.$$

Here  $p(t - s) \leq 0$  and  $s + 2t(1 - s) \leq s + (1 - s) = 1$ . If  $t \geq 1/2$ , then

$$(2t - 1)T_2 + 2(1 - t)T_3 = t(3 - 2t) + p(t - s) \leq t(3 - 2t) \leq \frac{9}{8}.$$

Again  $p(t - s) \leq 0$ , and the displayed coefficients are nonnegative and sum to one.

Thus a value above  $9/8$  can occur only in the region

$$0 < s \leq t \leq 1, \quad s + \frac{s}{t} \geq 1,$$

and there it is at most  $g(s, t)$ . Equivalently, optimize on the compact set

$$\overline{\mathcal{D}} := \{(s, t) : 0 \leq s \leq t \leq 1, s(t + 1) \geq t\}.$$

The only point with  $t = 0$  is  $(0, 0)$ . Defining  $g(0, 0) = 0$  makes  $g$  continuous there, because  $0 \leq s^2/t \leq s$  whenever  $0 < s \leq t$ . The unique interior stationary point of  $g$  satisfies

$$s = t^2, \quad 2t^2 + 3t - 4 = 0,$$

so

$$t = \frac{\sqrt{41} - 3}{4}, \quad s = t^2, \quad g(s, t) = L.$$

On the three nontrivial boundary pieces,

$$\begin{array}{ll} s = t & g = t(3 - 2t) \leq 9/8, \\ t = 1 & g = s(3 - 2s) \leq 9/8, \\ s + s/t = 1 & g = \frac{t(3 - t)}{t + 1} \leq 1. \end{array}$$

(The remaining boundary point is  $(0, 0)$  and has value zero.) Since  $L > 9/8$  (equivalently  $263 > 41\sqrt{41}$ ), the global maximum is  $L$ . Therefore every instance in the restricted model has a cost-feasible routing with overload at most  $L$ , proving the upper bound. The rational family from theorem 6 approaches  $L$ , so imposing rational data does not change the supremum.  $\square$

*Remark 9.* Theorem 8 does not cover unequal full expensive-route costs, a different cost threshold requiring three or four cheap choices, or other four-terminal topologies. Those remain separate optimization questions.



## 5 A finite integer certificate: 335/294

Choose

$$q = \frac{6}{7}, \quad \varepsilon = \frac{1}{10584},$$

so that

$$(p_1, p_2, p_3, p_4) = \left( \frac{13}{49}, \frac{97}{216}, \frac{1}{7}, \frac{1}{7} \right), \quad \sum_i p_i = 1 + \frac{1}{10584}.$$

Scale all demands and path amounts by 294. The resulting integer data are

| $i$ | $d_i$ | cheap amount | expensive amount |
|-----|-------|--------------|------------------|
| 1   | 294   | 78           | 216              |
| 2   | 216   | 97           | 119              |
| 3   | 252   | 36           | 216              |
| 4   | 294   | 42           | 252              |

and  $d_{\max} = 294$ .

Let  $M = 10584$ . Put per-unit costs

$$36 = \frac{M}{294}, \quad 49 = \frac{M}{216}, \quad 42 = \frac{M}{252}, \quad 36 = \frac{M}{294}$$

on  $(s, t_1), (s, t_2), (v_1, t_3), (v_2, t_4)$ , respectively, and zero on every other arc. Thus each full expensive choice costs  $M$ . The fractional cost is

$$\begin{aligned} \text{cost}_c(x) &= 216(36) + 119(49) + 216(42) + 252(36) \\ &= 31751 = 3M - 1 < 3M = 31752. \end{aligned}$$

Therefore every cost-nonincreasing unsplittable routing has at least two cheap choices.

For completeness, the fractional load on every arc is

| arc   | $x(a)$ | arc          | $x(a)$ |
|-------|--------|--------------|--------|
| $a_1$ | 721    | $(s, t_1)$   | 216    |
| $a_2$ | 505    | $(v_3, t_1)$ | 78     |
| $a_3$ | 253    | $(s, t_2)$   | 119    |
| $a_4$ | 175    | $(v_5, t_2)$ | 97     |
| $a_5$ | 133    | $(v_1, t_3)$ | 216    |
|       |        | $(v_5, t_3)$ | 36     |
|       |        | $(v_2, t_4)$ | 252    |
|       |        | $(v_4, t_4)$ | 42     |

Flow conservation at the nonterminals follows from

$$\begin{aligned} 721 + 216 + 119 &= 1056 = 294 + 216 + 252 + 294, \\ 721 &= 505 + 216, \quad 505 = 253 + 252, \\ 253 &= 175 + 78, \quad 175 = 133 + 42, \quad 133 = 97 + 36. \end{aligned}$$

The six exactly-two-cheap cases are

| $S$        | witness arc | $\text{flow}_{y^S}(a) - x(a)$ |
|------------|-------------|-------------------------------|
| $\{1, 2\}$ | $a_1$       | 335                           |
| $\{1, 3\}$ | $a_2$       | 335                           |
| $\{1, 4\}$ | $a_3$       | 335                           |
| $\{2, 3\}$ | $a_5$       | 335                           |
| $\{2, 4\}$ | $a_4$       | 335                           |
| $\{3, 4\}$ | $a_4$       | 371                           |

Together with lemma 4, this covers all eleven cost-feasible routings. Hence this single integer instance proves

$$\alpha_{\text{pl}} \geq \frac{335}{294} = 1.139455782312\dots$$

The lower bound is attained as the exact minimum maximum overload among the cost-feasible routings. For example, the routing with cheap set  $\{1, 2\}$  has full deviation vector

$$\begin{aligned} (a_1, \dots, a_5) &= (335, 299, 257, 41, 83), \\ ((s, t_1), (v_3, t_1)) &= (-216, 216), \\ ((s, t_2), (v_5, t_2)) &= (-119, 119), \\ ((v_1, t_3), (v_5, t_3)) &= (36, -36), \\ ((v_2, t_4), (v_4, t_4)) &= (42, -42). \end{aligned}$$

Its maximum deviation is exactly 335. Positive deviations on private terminal arcs are at most  $252 < d_{\text{max}}$ .

*Remark 10* (Binary arc costs). If costs in  $\{0, 1\}$  are preferred, replace the four charged private arcs of cost 36, 49, 42, 36 by directed paths of the corresponding lengths, with unit cost on every replacement arc. This preserves acyclicity, planarity, the two-route structure, all path costs, and the certificate.

## 6 What is and is not proved

The radical  $L$  is the exact maximum of the one-variable limiting function  $q^2(4 - q^2 - 2q)$  and, by theorem 8, the exact supremum for the specified equal-full-cost two-cheap model on the fixed topology. We do *not* claim that it is the exact value of  $\alpha_{\text{pl}}$ , the exact unrestricted four-terminal constant, the optimum on the fixed topology with unequal full-route costs, or the best four-terminal planar gadget. The result is also not an algorithmic lower bound in any special sense: it is an existential obstruction, and therefore applies a fortiori to any algorithmic version of the constant.

The supplementary verifier is not needed for the proof. It reconstructs the two paths per terminal from the edge list, checks connectedness and that a supplied rotation system defines a genus-zero embedding, verifies acyclicity, flow conservation, and legal arc costs, and enumerates all 16 unsplittable routings on all 13 arcs using exact integer arithmetic. Separate symbolic scripts derive the path-difference supports from the graph, verify the family identities and radical optimization, and audit the convex-combination proof of theorem 8. Mutation tests confirm that representative changes to the graph, amounts, costs, rotation system, minor certificate, and expected overload are rejected.

## 7 Research provenance and contemporaneous context

The investigation was directed by Matthew Protti. OpenAI GPT-5.6 Pro was used substantively for construction search, symbolic derivation, proof development, verifier development, and drafting. Separate model sessions and a Codex execution session reimplemented and stress-tested key claims from the stated graph and numerical data. These are computational cross-checks, not independent human verification or peer review. The author reviewed the claims, selected the scope of this disclosure, and accepts responsibility for the released work.

The investigation began after Dmitry Rybin’s public post of 22 July 2026 announcing a counterexample to Goemans’ cost conjecture. Because related work is developing rapidly, the novelty statements accompanying this note are expressly limited to targeted searches. Private development prompts and conversation transcripts are not released; the mathematical proof, explicit instance, and executable checks are intended to make the claims auditable without them.

## References

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## A The six symbolic substitutions

For auditability, here are the witness calculations without omitted steps. Let

$$b_2 := q^2(q^2 + 2q - 2 + \varepsilon).$$

Then

$$\begin{aligned} \{1, 2\}, a_1 : \quad & q^2 + q^2(3 - q^2 - 2q - \varepsilon) = R, \\ \{1, 3\}, a_2 : \quad & q^2 - b_2 + q^2 = R, \\ \{1, 4\}, a_3 : \quad & q^2 - b_2 + (q^2 - q) + q = R, \\ \{2, 3\}, a_5 : \quad & q^2(3 - q^2 - 2q - \varepsilon) + q^2 = R, \\ \{2, 4\}, a_4 : \quad & q^2(3 - q^2 - 2q - \varepsilon) + (q^2 - q) + q = R, \\ \{3, 4\}, a_4 : \quad & -b_2 + q^2 + q = R + q - q^2. \end{aligned}$$